

4(a)(i)	oscillations are in a single direction, which is perpendicular to the direction of propagation (of the wave) or oscillations are in a single plane, which contains the direction of propagation (of the wave)	B1
4(a)(ii)	light waves are transverse and sound waves are longitudinal	B1
4(b)	$I = I_0 \cos^2 \theta$	C1
	$\cos^2 \theta = 1/4$ so $\cos \theta = 1/2$ $\theta = 60^\circ$ or 120° or 240° or 300°	C1
	angle of rotation = $(120^\circ - 90^\circ)$ or $(240^\circ - 90^\circ)$ or $(300^\circ - 90^\circ)$ $= 30^\circ$ or 150° or 210° or 330°	A1
4(c)(i)	the waves have different amplitudes	B1
	cannot have resultant displacement that is always zero or cannot have (complete) destructive interference (at nodes) or (at nodes resultant) amplitude is the difference of the amplitudes	B1
4(c)(ii)	$I \propto A^2$	C1
	$A^2 / A_0^2 = (I_0 / 4) / I_0$ $A = 0.5 A_0$	C1
	maximum amplitude = $A_0 + 0.5 A_0$ $= 1.5 A_0$	A1